

# YASH KUMAR

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## PROFESSIONAL SUMMARY

B.Tech AI/ML student at GGSIPU (2024–2028) with hands-on experience in computer vision, deep learning, and real-time object detection. Built and deployed production-ready ML systems using YOLOv8, OpenCV, and PyTorch, and developed an AI-powered study agent on Google Cloud Gen AI infrastructure. Proficient in LangChain, RAG pipelines, and LLM integration, with full-stack skills (React, Next.js, FastAPI) enabling end-to-end deployment of AI-powered applications.

## EDUCATION

### B.Tech in Artificial Intelligence and Machine Learning

University School of Automation and Robotics, GGSIPU EDC

2024 – 2028 (Expected) **CGPA: 8.0+**

**Relevant Coursework:** Data Structures & Algorithms, Machine Learning, Artificial Intelligence, DBMS, Operating Systems, Computer Networks, OOPs, Computer Vision

## TECHNICAL SKILLS

**Languages:** Python, C++, JavaScript (ES6+), TypeScript, SQL  
**ML / DL:** PyTorch, TensorFlow, Keras, Scikit-learn, YOLOv8, OpenCV  
**AI & Data:** LangChain, LangGraph, RAG, ChromaDB, NumPy, Pandas, Matplotlib, Google Cloud AI  
**Frontend:** React.js, Next.js, Tailwind CSS, Redux, Streamlit  
**Backend & DB:** FastAPI, Flask, Node.js, Express.js, PostgreSQL, MongoDB, Firebase  
**Tools:** Git, GitHub, Docker, Kubernetes, Vercel, Postman, Linux

## KEY PROJECTS

### QueDB — AI-Powered Natural Language to SQL Engine [github.com/Yashcodes07/QueDB](https://github.com/Yashcodes07/QueDB)

*Tech Stack: FastAPI, React, PostgreSQL, LangChain, Groq (LLaMA-3 70B), ChromaDB, SQLAlchemy, JWT Auth*

- Engineered a full-stack AI query engine that converts natural language to executable SQL across multiple database types (PostgreSQL, MySQL, SQLite) using LLaMA-3 70B via Groq API — enabling non-technical users to query databases without writing SQL.
- Built a RAG-based knowledge base using ChromaDB vector store and sentence-transformers, allowing users to store domain context that the LLM references for more accurate, schema-aware query generation.
- Implemented multi-retry SQL execution with error feedback loop — if generated SQL fails, the error is passed back to the LLM for self-correction, achieving higher query success rates automatically.
- Designed a live Schema Explorer that introspects connected database structure (tables, columns, types, relationships) and displays it in an interactive tree view alongside a Monaco code editor for query editing.

### Study Guide AI Agent (Temporarily offline - migrating Hosting)

*Tech Stack: Google Cloud, Gen AI APIs, Python, Cloud Run, LLM Integration, ADK*

- Built and deployed an AI-powered study guide agent as part of Google Cloud Gen AI Academy APAC 2026 competition using Google's Agent Development Kit (ADK).
- Leveraged Google Cloud Run for scalable, production-ready serverless deployment on cloud infrastructure.
- Integrated LLM-based capabilities to generate structured, topic-specific study content dynamically for students.

### Helmet Detection & Traffic Monitoring System [github.com/Yashcodes07/Helmet-Detection-Project](https://github.com/Yashcodes07/Helmet-Detection-Project)

*Tech Stack: Python, YOLOv8, OpenCV, FastAPI, NumPy, HTML/CSS, JavaScript*

- Built a real-time helmet detection system using deep learning to classify helmet vs. no-helmet riders from live images and video streams via a FastAPI REST backend with auto-generated Swagger docs.
- Implemented YOLO-based object detection pipeline with overlap-based person↔motorcycle matching; achieved mAP@0.5: 94.6% and mAP@0.5:0.95: 73.9% on test dataset.

- Achieved inference speed of ~5ms per image; designed multi-stage detection pipeline with base64 image response and video processing support.

**AkashSan Cleaning — Business Website** [akashsan.in](https://akashsan.in)

*Tech Stack: Next.js, React, Tailwind CSS, TypeScript, SEO Optimization, Vercel*

- Designed and deployed a full production business website for a Delhi NCR cleaning company — responsive UI, SEO-optimized with meta tags, structured data, and Google Search Console integration.
- Implemented multi-service landing page with WhatsApp booking integration, testimonials, and mobile-first responsive layout.

**Dynatech Controls — Official Company Website** [dynatechcontrols.in](https://dynatechcontrols.in)

*Tech Stack: Next.js, React, Tailwind CSS, TypeScript, Resend API, Vercel*

- Designed and delivered the full production website for Dynatech Controls — including build, hosting, and domain integration with contact form via Resend API.
- Built responsive, SEO-optimized UI using Next.js and Tailwind CSS, optimized for desktop, tablet, and mobile.

**CERTIFICATIONS**

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|------------------------------------------------------------------------------|----------|
| <b>IBM AI Fundamentals   IBM SkillsBuild</b>                                 | May 2026 |
| <b>Google Cloud Gen AI Academy APAC 2026   Google Cloud &amp; Hack2skill</b> | Apr 2026 |
| <b>Deloitte Data Analytics Job Simulation   Forage</b>                       | Mar 2026 |
| <b>JPMorgan Chase Software Engineering Job Simulation   Forage</b>           | Mar 2026 |

**ACHIEVEMENTS & ACTIVITIES**

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- SIH 2025 — Certificate of Appreciation; cleared University-Level 2 Rounds of Smart India Hackathon.
- Google Cloud Gen AI Competition — Built and deployed a live AI agent on Google Cloud Run as part of APAC 2026 Cohort 1.
- CodeX Hackathon — Participated and developed a project under competitive hackathon conditions.